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Full-stack software engineer and designer with over 7 years of experience. Across engineering and executive leadership roles, I've led technical teams, advised CTOs, served as board president of a historic building, and specialized in technical, legislative, legal, compliance, regulatory, utility, and financial research. I like making things that are useful to people and that work well.

EMPLOYMENT

Quorum Analytics		Washington, DC
2022–2024	Senior Software Engineer	(50 to 70+ hours/week)
2020–2022	Software Engineer II	
2018–2020	Software Engineer	

Python, Django, SQL, PostgreSQL, JavaScript, TypeScript, React, Redux, D3, HTML, CSS, React Native, PostGIS, Elastic

Wrote and launched over a dozen flagship products for Quorum Analytics, the industry leading public affairs platform. I consistently solved the team’s most complex technical challenges while delivering projects on time, in scope, and under budget.

- Instrumental in helping scale ARR from \$9M to \$64M+.
- Led cross-functional civic tech projects in collection, organization, and presentation of legislative, regulatory, compliance, financial, geospatial, and user-generated datasets.
- Authored 5k+ commits, 1k+ code reviews, 700+ PRs, and conducted 100+ technical interviews.

Projects			Duties	Leadership
PAC Hybrid	Technical Lead	Team of 10	Mentorship, coaching, onboarding	Wrote first design system
Quorum PAC	Technical Lead	Team of 8	Code Review (async, collaborative)	Created full-stack guild
Search UI/UX	Technical Lead	Team of 4	Interviews (technical, design)	Led apprenticeship program
Dashboards	Technical Lead	Team of 2	Tests (regression, integration, unit)	Mentored dozens of engineers
Transcripts	Frontend Lead	Team of 2	Technical Design Documents	Onboarding coach
Bulk Actions	Technical Lead	Team of 1	Product Requirements Documents	Upgraded major dependencies
International	Full-stack	Team of 2	Technical Documentation	Discovered bug in Google Chrome
Regulations	Full-stack	Team of 1	“Learning Lab” presentations	Longest tenured engineer
Search	Full-stack	Team of 2	Deployment, 24hr on-call shifts	Rockstar of the Quarter

Massachusetts Institute of Technology: Lincoln Laboratory (MIT LL)		Lexington, MA
Summer 2016	Research Intern	(40 hours/week)

Java, Android SDK, Google Maps SDK, JavaScript, ExtJS, HTML, CSS, SQL, Acme (Plan 9)

Wrote Local Evacuation Alert Verification (LEAV, sponsored by DHS S&T) for HURREVAC-eXtended (HVX, sponsored by FEMA) with the Humanitarian Assistance and Disaster Relief Systems (HADRS) group at MIT LL on Hanscom AFB.

- Designed and built LEAV, an Android application that routes hurricane evacuees to shelters, visualizes evacuation zone status in an interactive map, and facilitates communication between Emergency Managers (EMs) and evacuees.
- Developed an interactive LEAV web module which enables Emergency Managers to update, visualize, and monitor the status of state evacuation zones in the HVX web interface.
- Reduced WFS-T request size by an order of magnitude.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Go, SQL, LaTeX	SCM & CI/CD	Git, Github, Jenkins, Travis
Databases	PostgreSQL, PostGIS, Elasticsearch, SQLite	Infrastructure	Docker, Kubernetes, Ansible
Web	Django, React, Next.js, D3, HTML5, CSS3	Observability	Sentry, Datadog, Grafana
Devops	AWS (EC2, RDS, S3, Lambda, DMS, Route53)	Design	Figma, Invision, Sketch, GIMP
	Atlassian (Jira, Confluence, Opsgenie), G Suite	Unix	sh, pkg-src, brew, x11, plan9port
Libraries	npm, Babel, Webpack, Gulp, Cypress, Mantine, Styled Components, Stylus, Jest, react-testing-library		

EDUCATION

Gettysburg College	Gettysburg, PA
Bachelor of Science in Computer Science (3.60, HONORS) and Philosophy (3.73, HONORS)	2013–2017
Class of 2017 Outstanding Computer Science Student Award	

PERSONAL PROJECTS

In my free time, I enjoy working on creative and idiosyncratic projects. Recently scraped and analyzed the Electronic Code of Federal Regulations and extended bob—a Golang query builder—with Django features and syntax. Previously ported James Ashton’s X-Face decoder from C to Golang, wrote wiki and blog platforms in Golang, found a bug in the Google Chrome URL parser, discovered a novel method to losslessly convert Susan Kare’s Macintosh fonts from Classic Mac OS resource forks to macOS data forks, wrote a JavaScript weather script with 90’s bitmaps, created a 3x4 bitmap font, wrote a Mozilla add-on to display ‘face’ images embedded in email and newsgroup message headers, and won a Gentoo Linux interface design competition. My personal, professional, and open-source work is detailed in my full CV: <https://johnduncan.io/cv.pdf>.

PROFESSIONAL PROJECTS

Quorum Analytics

Washington, DC

PAC Hybrid | Python, Django, SQL, TypeScript, React, Mantine, K8s, PostgreSQL, MySQL, AWS DMS, Figma 2024
As technical lead, designed, implemented, and launched the Quorum PAC Hybrid FEC reporting platform, further expanding Quorum into the \$15M+ PAC software market by integrating Cision's government relations PAC product into Quorum's federal PAC platform. Wrote majority of backend queries, full-stack search infrastructure, technical design document (TDD) for one-time and recurring credit card contributions, and TDD and MVP for the full-stack Dashboard integration. Reduced database query latency from 60 seconds to 20 milliseconds and backfill execution time from two weeks to under five seconds.

Quorum PAC | Python, Django, FastAPI, TypeScript, React, Mantine, Docker, K8s, PostgreSQL, Figma 2022–2024
As technical lead, designed and implemented full-stack search and dashboard infrastructure for the Quorum PAC FEC reporting platform, a greenfield project expanding Quorum's business into the \$15M+ PAC software market and helping clients remain compliant, raise funds, and report results to the FEC.

Search UI/UX | Python, Django, JavaScript, React, Styled Components, PostgreSQL, Invision 2021
As technical lead, designed and implemented UI/UX enhancements for Quorum's search feature, the default homepage for users. Wrote a majority of full-stack project infrastructure, collaborated with product design team on the product requirements document (PRD), wrote the project technical design document (TDD), and wrote dozens of backend database queries.

Dashboards | Python, Django, SQL, JavaScript, React, D3, Styled Components, Postgres, PostGIS, Invision 2020
As technical lead, developed an industry-first dashboard feature empowering users to visualize all of Quorum's data in an intuitive, centralized interface. Wrote full-stack frontend and backend infrastructure, over a hundred backend database queries, and all frontend components, including core react-grid-layout dashboard, widgets, visualizations, and maps.

Transcripts | Python, Django, JavaScript, React, PostgreSQL, AWS S3/RDS 2019
Developed the frontend for the Committee and Presidential Transcripts feature, enabling users to read, search, download, and visualize transcripts of congressional hearings and presidential speeches within an hour of their completion.

Bulk Actions | Python, Django, SQL, JavaScript, React, PostgreSQL 2019
As technical lead, designed and wrote a full-stack Bulk Action feature enabling users to select, update, or delete up to 1 million rows at once across 17 datasets and 13 action types. Researched and wrote optimized batch create/update queries using Django ORM and raw SQL, culminating in over a dozen highly optimized O(1) SQL QueryMethods for scalable performance.

International | Python, Django, SQL, JavaScript, React, PostgreSQL 2018
Scaled Quorum product to new continents and dozens of new countries by writing the full-stack infrastructure for the international search feature, allowing clients to gather intelligence, identify trends, and communicate with key international stakeholders. Wrote backend database queries and ability to filter and visualize new international data in search datasets.

Regulations | Python, Django, SQL, JavaScript, React, PostgreSQL 2018
Wrote full-stack search and profile infrastructure for the regulations feature, which gives clients access to comprehensive state & federal regulatory data and tracking tools. Wrote dozens of frontend filters and complex backend database queries, integrated feature into existing Quorum alert, list, and download infrastructure, and added ability to set stances, priorities, and issues. Rewrote frontend Bill timeline to support Regulations and other arbitrary datasets in the future.

Search | Python, Django, SQL, JavaScript, React, HTML, PostgreSQL, PostGIS 2018
Wrote and expanded full-stack functionality for search feature, the default homepage for all Quorum clients. Wrote infrastructure for 30+ datasets, hundreds of new filters, a dozen new visualizations supported across all datasets, new interactive search results, and expanded support for csv downloads. Wrote shared infrastructure powering search on the desktop (React) and mobile (React Native) applications. Wrote a library which converts raw HTML into React Elements by recursively processing DOM nodes and converting them into React elements.

LEADERSHIP

The Cairo Board of Directors

2023–present

Served as The Cairo's (1894) interim general manager on contractual basis during transition between management companies.

- The Cairo is listed on the DC Inventory of Historic Sites (1990) and the National Register of Historic Places (1994).
- Developed an improved financial management plan after conducting a forensic audit of condo finances, optimizing expenditures, restructuring staff schedules to maintain employees without layoffs, enhancing vendor accountability, and implementing a proactive capital improvement strategy—all while maintaining fee increases below 5%. Drafted FY '26 budget whose 172k operating surplus allocates 64k toward capital improvements and 67.5k toward replacement reserves.
- Performed legal research resulting in \$40,000 settlement in Association's favor, discovered Pepco meter issue resulting in unbilled 265v HVAC charges for 11 months, discovered Pepco violated DC tariffs by charging 100% of deficiency in billing, and settled a \$100,000+ lawsuit from prior Board.
- Partnered with a horticulturist from Harvard's Dumbarton Oaks Gardens and a landscape architect from Oehme, van Sweden & Associates to design and install a perennial pollinator garden. Design integrates heirloom plants which are concomitant with period appropriate Victorian varieties.